Math for CS & AI: Homework 4

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(Adapted from earlier versions of this course taught by Andrew Yao)

Due Date: Mar 28, 2026

Problem Set

- Exercises 2.24–2.29 from Chapter 2 and 3.1–3.2 from Chapter 3 of the lecture notes: https://cloud.tsinghua.edu.cn/d/57ddec714cf74c1cab85/

Writing Guidelines

- Same as the first homework.
- We may deduct up to 3 points for poor writing.
- **Special Note:** Your submission should contain only a single PDF file. Please do not submit other files such as `*.zip` or `*.tex`.

Additional Hints (Experimental)

- We provide additional hints for some of the homework problems in the next few pages.
- If you get stuck on a problem, scroll down to find that problem and read the hints.
- Scroll *very slowly* so that you do not accidentally see hints for problems you have not thought about yet.
- Homework is meant to help you practice your problem-solving skills. It is your job to make sure you have thought enough before reading the hints. Ultimately, there will be no additional hints and no AI tools in the exams.
- Again, **peer learning is strongly encouraged**. Reading hints or getting help from AIs is not the same as learning from your classmate who has thought about the problem and can explain it to you. Make some friends who are willing to discuss problems or even have hotpot with you. Learn from their way of thinking, including both successful and unsuccessful attempts.

The rest of this document contains hints for the homework problems.



Think twice before scrolling down!

Hints

Problem 2.24

- Linearity of expectation.
- Compute $\Pr[\sigma_i > \sigma_j]$ for all $i < j$.

Problem 2.25

- There are many ways to construct such random variables.
- One way is to first construct n random variables such that $X_1 > X_2 > \cdots > X_n$ holds with probability 1.
- Then do some random thing.
- Best possible β? Please go thorough the properties of expectation we have learned so far.

Problem 2.26

- Do not use brute force.
- Try to rewrite the conditional expectation in a way that you can apply the law of total expectation.

Problem 2.27

No hint.

Problem 2.28

No hint.

Problem 2.29

- Do not overthink. The hash function is very simple.
- The following fact may be useful: to estimate the mean of n integers, one can take one or more samples and use their average as an estimator.

Problem 3.1

- Please go thorough the properties of expectation we have learned so far.

Problem 3.2

- For (a), show that
$$f(x) \geq \frac{1}{t}\left(f(tx + (1-t)y) - f(y)\right).$$
- For (b), use the inequality $f(x) \geq f(y) + \alpha(x - y)$ for some x, y and linearity of expectation.